

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Let's talk about computers development!



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.
Make the information system design report, based on the selection of both hardware and software tools required for the IT solution.

Ask your partner:

- How do computers store big amounts of data?
- Do you know the difference between a RAM and a hard drive?

Can Do Statements:

- I can talk about computation and computer architecture.
- I can talk about the history of computation and computer architecture.
- I can ask questions related to computation and computer architecture.





Let's Explore!

What were the most important contributions to technology?



Instructions: Read the text below.

Thomas: Hi, Linda! What are you doing?

Linda: Hey, Thomas! I'm reading this article about computation and computer architecture for my class. There is too much information about it. It says that John Von Neumann, a mathematician and computer scientist set up the main principles of computers architecture and design. He predicted too many features that modern computers have today, such as the data stored in the internal memory of machines, the use of a RAM memory or fast memory to hold commands, and the hard drive or slow memory to hold larger quantities of data.

Thomas: Wow! Tell me more!

Linda: Sure! Related to computation and computer architecture history, John Von Neumann immigrated to the United States in 1930. There, he got interested in programmable calculators invention. Years later, he started to work on aerodynamics and ballistics problems. As these problems required machine assistance, he had the opportunity to know the programmable calculator "Harvard Mark I". Later, the ENIAC appeared, an electronic digital computer that was faster than the Mark I. As you can see, there were too many prototypes of computers until the most recent versions nowadays.

Thomas: I have some questions about computation. What happened with the ENIAC?

Linda: Neumann took some ideas from the ENIAC to devise a design that could store information in a functional way with the binary system, likewise some marks that indicated the memory by voltage or magnetism.

Thomas: Who else made important contributions to computation?

Linda: Steve Jobs developed too many important ideas about it. He founded Apple Computer Company in 1976, which launched Lisa (GUI) a computer with a graphical user interface. Years later, he released the first computer with a mouse, the Macintosh. In 1998, Apple launched the iMac, a computer with access to the Internet.





Let's Practice!

Thomas is learning about computer architecture.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. Von Neumann's computer architecture
2. In computation, we define RAM memory
3. Von Neumann talked about the hard drive as a device

COLUMN B

- a. As an internal device that maintains available too many software commands.
- b. that can hold and recover too much information.
- c. described too many important characteristics of computers.



Let's Practice!

Important Information About Steve Jobs.

Instructions: Select the correct answer.

1. Steven Paul Jobs was born in
 - a. Nevada
 - b. San Francisco
 - c. Oregon
2. The first computer released by Jobs was:
 - a. Apple I
 - b. GUI
 - c. Macintosh
3. Jobs founded Apple Computer Company in:
 - a. 1972
 - b. 1976
 - c. 1983
4. The first computer with mouse was
 - a. Apple I
 - b. GUI
 - c. Macintosh
5. Jobs bought the enterprise The Graphics group in
 - a. 1986
 - b. 1996
 - c. 1998
6. Jobs died due to a
 - a. pancreatic cancer
 - b. skin cancer
 - c. diabetes





Extension Activity

Computation and Computer Architecture in Your Community.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Do some research about computer architecture.
- Write a dialogue where you explain the importance of computer architecture"
- Share with a partner.

Self-Assessment

Yes

No

1. I research about computer architecture.

☐
☐

2. I write a dialogue where I explain the importance of the computer architecture.

☐
☐

3. I share with a partner.

☐
☐


Target Words

- **internal** (adjective): located in the inner part of something
e.g. He predicted too many features that modern computers have today, such as the data stored in the internal memory of machines.
- **RAM memory** (noun): (Random Access Memory) it is a device used to store data which let read or write these ones in a faster way
e.g. He predicted too many features that modern computers have today, such as the data stored in the internal memory of machines, the use of a RAM memory or fast memory to hold commands.
- **aerodynamics** (noun): the analysis of air or other gases motion
e.g. Years later, he started to work on aerodynamics and ballistics problems.
- **programmable calculator** (noun): device that allows you to make calculations, the first computers were programmable calculators
e.g. As these problems required machine assistance, he had the opportunity to know the programmable calculator "Harvard Mark I".
- **prototype** (noun): a model, a design
e.g. As you can see, there were too many prototypes of computers until the most recent versions nowadays.
- **magnetism** (noun): something with the property of attraction due to a magnet
e.g. Neumann devise a design that could have some marks that indicated the memory by voltage or magnetism.



Learning Points

Quantifiers: Too Many and Too Much.

You use **quantifiers** to express the quantity of something. In this order, quantifiers modify nouns in terms of amount.

Two common **quantifiers** are: ***Too many*** and ***too much***.

You use ***too many*** with countable nouns in plural.

Example:

- There are ***too many*** programs for each specific task in the computer.

You use ***too much*** with uncountable nouns in singular (due to their condition of uncountable nouns there's no plural form for them).

Example:

- He has ***too much*** experience in computer architecture.

(CHART 1)

Too Many (Countable nouns)	
Singular	Plural
• <i>Program</i> • <i>Computer</i> • <i>Field</i> • <i>Design</i>	• <i>Programs</i> • <i>Computers</i> • <i>Fields</i> • <i>Designs</i>

(CHART 2)

Too Much (Uncountable nouns)
Singular
• <i>Experience</i> • <i>Performance</i> • <i>Information</i> • <i>Time</i>

Examples

- *He predicted too many features that modern computers have today, such as the data stored in the internal memory of machines.*
- *As you can see, there were too many prototypes of computers until the most recent versions nowadays.*
- *Steve Jobs developed too many important ideas about it.*





Learning Points

Steve Jobs.

Steve Jobs is one of the most important genius in the computing technology.

He was the inventor of the first personal computer and the founder of Apple Computer Company. He was a visionary that worked very hard to take advantage of technology transforming it into entertainment for users around the world.

Check this brief profile about his life:

STEVE JOBS (1955-2011)	
1955	Steven Paul Jobs was born in San Francisco, United States. He was adopted by Paul and Clara Jobs.
1972	He got into The Reed College University in Portland (Oregon). He studied only for six months due to the high costs of his studies.
1976	He founded Apple Computer Company. In ten years, this company got 4000 employees. The first computer was the Apple I, considered as the first personal computer.
1983	Apple released Lisa (GUI), another personal computer that offered a graphical user interface. This project was unsuccessful due to the high cost of this computer in that time.
1984	Jobs and his friend Steve Wozniak released the Macintosh computer. It was the first computer with mouse.
1985	Jobs left Apple due to some disagreements with John Sculley, the executive director of his own company. Sculley was in other times the executive director of Pepsi-Cola and Jobs persuaded him to work for Apple, with the intention to keep the competitiveness of his company. This year he founded NeXT Computer.
1986	Jobs bought the enterprise The Graphics group, which will become Pixar later.
1996	Apple bought NeXT Computer, which meant the return of Jobs to the company.
1998	Apple released the iMac, a computer designed to surf on the Internet.
2001	Jobs released the iPod, a portable media player.
2007	Jobs released the iPhone, a smart phone that revolutionized the concept of entertainment in mobile technology.
2011	Jobs died due to a pancreatic cancer that was diagnosed in 2003.